

BULGAN BURENKHAIKHAAN, Mongolia

WMO: 442650

Lat: 46.0947N

Lon: 91.5522E

Elev: 3891

StdP: 12.74

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.2	-24.2	-33.5	1.0	-28.6	-28.8	1.3	-23.7	13.9	7.6	11.0	11.9	0.3	50	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	89.3	63.6	86.4	62.4	83.7	61.5	67.9	83.7	65.6	81.6	63.6	79.1	5.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	96.1	76.8	59.6	88.1	74.7	57.4	81.1	73.7	34.7	83.7	32.7	81.7	31.0	79.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.1	13.8	-34.9	94.9	7.8	3.0	-40.5	97.0	-45.1	98.7	-49.5	100.4	-55.1	102.5		
(5)			-34.7	69.4	8.1	4.3	-40.5	72.5	-45.2	75.0	-49.8	77.4	-55.6	80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	-4.2	5.3	25.7	45.4	57.3	67.8	70.9	67.1	55.3	40.1	19.4	2.0		
	DBStd	27.62	10.01	10.98	12.54	9.35	7.27	5.61	4.75	5.43	7.34	8.36	12.12	9.69		
	HDD50	6649	1679	1250	755	187	21	0	0	0	32	318	918	1489		
	HDD65	10348	2144	1670	1220	589	252	35	7	39	297	773	1368	1954		
	CDD50	2211	0	0	0	48	249	534	648	531	192	9	0	0		
	CDD65	436	0	0	0	0	15	119	190	105	7	0	0	0		
	CDH74	5043	0	0	0	8	250	1284	2015	1325	161	0	0	0		
	CDH80	1565	0	0	0	0	39	372	719	412	23	0	0	0		
(14) Wind	WSAvg	3.6	1.2	1.7	3.4	5.5	5.8	5.5	4.8	4.2	3.9	3.3	2.1	1.4		
(15) Precipitation	PrecAvg	3.3	0.2	0.1	0.2	0.2	0.2	0.5	0.6	0.5	0.2	0.2	0.3	0.2		
	PrecMax	5.5	1.1	0.6	0.6	0.8	1.4	1.8	1.5	2.8	0.6	0.7	1.5	0.7		
	PrecMin	1.7	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.0	0.2	0.1	0.2	0.2	0.3	0.4	0.4	0.6	0.2	0.2	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	37.4	61.0	75.1	84.6	90.8	95.1	91.4	83.3	69.6	52.7	30.6		
		MCWB	20.3	28.6	44.2	54.1	59.5	64.9	68.2	64.6	60.5	52.7	40.5	23.9		
	2%	DB	20.3	31.8	55.5	70.0	80.2	87.4	90.3	87.9	78.6	64.4	46.7	24.8		
		MCWB	17.4	26.5	41.5	51.0	57.6	61.8	65.1	62.9	58.2	48.5	36.8	20.4		
	5%	DB	16.0	27.3	50.6	66.1	76.6	84.4	87.0	84.7	74.7	60.5	41.9	20.9		
		MCWB	13.6	23.4	39.0	49.0	56.1	61.2	63.3	62.0	55.9	46.5	33.8	17.8		
	10%	DB	11.7	22.9	45.6	62.5	72.8	81.5	84.0	81.5	71.2	56.1	36.8	16.9		
		MCWB	9.9	19.9	36.2	47.0	53.9	60.1	62.1	60.5	54.2	44.0	30.5	14.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	30.4	45.9	56.1	63.6	69.3	71.8	69.1	65.4	55.1	41.5	24.6		
		MCDB	24.5	34.8	57.7	72.4	81.1	87.2	89.8	85.2	79.3	67.7	50.9	29.5		
	2%	WB	17.5	26.9	42.5	52.6	60.2	65.6	68.8	66.6	60.4	50.1	37.8	20.8		
		MCDB	19.9	31.4	54.2	67.4	76.4	82.7	85.7	82.4	74.7	62.1	46.7	24.3		
	5%	WB	13.6	23.4	39.7	50.2	57.4	63.5	66.5	64.3	57.5	47.3	34.4	18.0		
		MCDB	15.9	27.3	50.1	63.9	73.5	79.6	82.2	80.8	71.3	58.4	42.0	21.0		
	10%	WB	9.9	19.8	36.3	47.9	55.1	61.6	64.1	62.1	54.9	44.5	31.2	14.8		
		MCDB	11.7	22.8	44.9	61.0	70.7	76.6	78.9	77.7	69.3	55.3	37.3	17.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.1	26.2	24.6	23.6	22.6	21.7	22.2	24.4	26.5	25.5	22.4	21.7		
		MCDBR	25.0	27.3	28.5	29.8	28.3	27.0	27.9	29.5	31.0	32.1	28.9	24.0		
		MCWBR	22.2	22.2	17.7	16.8	14.5	12.3	12.3	13.3	16.0	19.9	19.4	19.8		
	5% WB	MCDBR	24.8	26.4	27.7	27.7	25.8	22.9	25.1	26.6	28.7	30.6	27.0	23.4		
		MCWBR	22.0	21.8	17.5	16.5	14.8	12.7	13.6	14.3	16.9	20.2	19.0	19.6		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.298	0.339	0.385	0.379	0.390	0.381	0.359	0.330	0.317	0.286	0.267		
	taud		2.271	2.187	2.154	2.117	2.209	2.254	2.315	2.356	2.387	2.340	2.317	2.251		
	Ebn at Noon		266	275	279	275	281	278	279	281	281	266	254	249		
	Edh at Noon		25	34	41	46	44	42	39	36	32	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		625	967	1418	1840	2174	2237	2184	1951	1614	1057	646	502		
	RadStd		31	37	88	101	126	138	91	65	77	59	52	31		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page